

REVIEW
for
FINAL EXAM

Chapters

- Introduction to Data Structures and Algorithms
 - Course slides: [07_Intro_DataStructures.pdf](#)
 - zyBooks: Ch. 6. Introduction to Data Structures and Algorithms
- Algorithm Analysis
 - Course slides: [08-AlgorithmAnalysis.pdf](#)
 - zyBooks: Ch. 8. Searching and Algorithm Analysis
- Searching and Algorithm Analysis
 - Course slides: [09-SearchingAndAlgorithmAnalysis.pdf](#)
 - zyBooks: Ch. 8. Searching and Algorithm Analysis
- Abstract Data Types (ADT)
 - Course slides: [10-ADTs and Lists.pdf](#)
 - zyBooks: Ch. 6. Introduction to Data Structures and Algorithms

Chapters

- **Graphs**

- **Course slides:**

- 11-Graphs.pdf

- 11-demo-dag.pdf

- zyBooks: Ch. 13. Graphs

- **Binary Search and MergeSort Algorithms**

- **Course slides:**

- 12-Binary Search and MergeSort Algorithms.pdf,

- 12-demo-mergeSort

- zyBooks: Ch. 14. Sorting Algorithms

Chapters

- Shortest Paths in a Graph (Dijkstra's Algorithm)
 - Course slides:
 - 13-Dijkstra-Shortest Path Algorithm.pdf
 - 13-demo-dijkstra
 - zyBooks: Ch. 13. Graphs
- Recursion and MasterTheorem
 - Course slides: 14-Recursion and MasterTheorem.pdf
 - zyBooks: Ch. 8. Searching and Algorithm Analysis

Chapters

- Minimum Spanning Trees
 - Course slides:
 - 15-mst.pdf
 - 15-demo-mst.pdf
 - zyBooks: Ch. 13. Graphs
- QuickSort
 - Course slides: 16-QuickSort.pdf
 - zyBooks: Ch. 14. Sorting Algorithms
- Binary Trees
 - Course slides: 17-Binary-Trees.pdf
 - zyBooks: Ch. 10. Trees

Chapters

- Huffman Coding
 - Course slides: 18-huffman.pdf
 - zyBooks: Ch. 13. Graphs
- AVL Trees
 - Course slides: 19-AVL Trees.pdf
 - zyBooks: Ch. 11. AVL Trees
- Heaps and Treaps
 - Course slides: 20-HeapsTreaps.pdf
 - zyBooks: Ch. 12. Heaps and Treaps

Chapters

- List ADT

- Course slides: 21-ListADT.pdf
 - zyBooks: Ch. 7. Lists, Starcks, and Queues

- Tries

- Course slides: 22-Tries.pdf
 - zyBooks: Ch. 10. Trees

- Hash Tables

- Course slides: 23-HashTables.pdf
 - zyBooks: Ch. 9. Hash Tables

Question Types

- Comprehensive exam (Topics after midterm exam)
 - At least one question from each chapter
- True / False (with explanation) questions
- Multiple Choice questions
- Matching questions
- Questions similar to homework questions in test-like format (Multiple Choice)